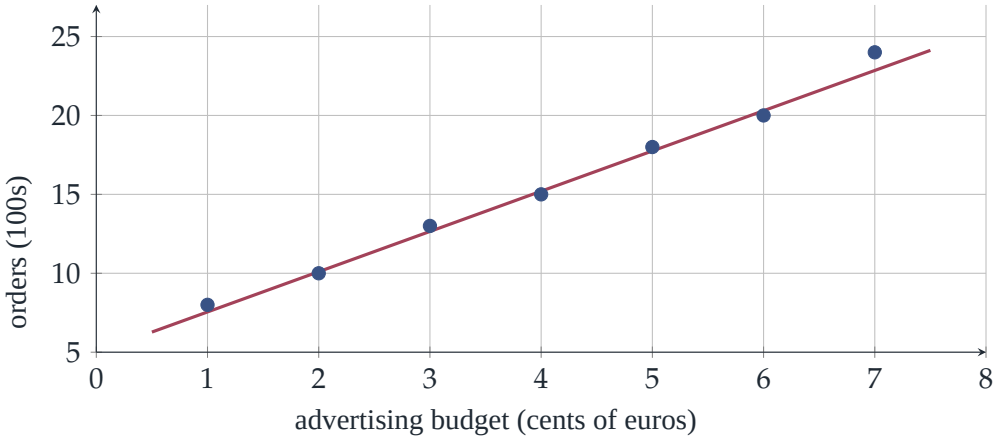
Chapter 5

Statistics with two quantitative variables

Competences: represent, choose an adjustment, interpolate, extrapolate with caution, communicate

5.1 Reference course

A two-variate statistical series consists of couples (x_i, y_i). The cloud of points allows to observe a trend, dispersion and possible atypical values.



Key ideas

A straight line of adjustment $y = ax + b$ provides estimates, not exact equities.
a described the estimated mean change in y when x increases by one unit.
remains in the range of the observed values; extrapolation emerges from this interval and must be presented with caution.

The average point $G(\bar{x}, \bar{y})$ a for coordinates $\bar{x} = \frac{x_1 + \cdots + x_n}{n}$ and $\bar{y} = \frac{y_1 + \cdots + y_n}{n}$. A right of regression obtained at the calculator passes through G, to the nearest round.

Method

To exploit $y = ax + b$: 1) replace x by the value requested to estimate y ; 2) resolve $ax + b = y_0$ to estimate x ; 3) round according to context; 4) specify interpolation or extrapolation ; 5) to remind that the result depends on the model.

A change in variable, for example $z = \log(y)$ or $z = \frac{1}{y}$, can make a cloud approximate- We'll adjust it then. (x, z), then we go back to the variable y.

Meeting 31 — Dot cloud

Two-variate statistics

26/01/2027 • B : 2 h – C : 1 h

Objectives of the session

· Build a cloud.

· Choose scales.

· Identify variables.

Fluency · 6 to 8 minutes, without calculator

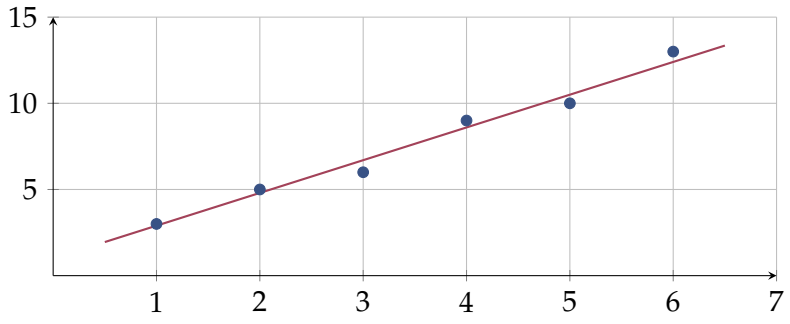
1. Write the multiplier coefficient of a 10 per cent increase. $1+\frac{10}{100}=1,10$
2. Resolve 3 $x+4=25$. $x=7$
3. Calculate the average of 6, 8 and 10. 8
4. Give the slope between $(1,3)$ and $(4,9)$. $\frac{9-3}{4-1}=2$
5. With $y=2x+5$, calculate y for $x=7$. 19

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today.
Before the activity, reformulate the keywords: build a cloud and choose the scales.

Research activity

The table (x_i, y_i) gives the Expenditure advertising and the sales of Six months : $(1,12), (2,15), (3,17), (4,22), (5,24), (6,28)$. Choosing scales, plotting the cloud and placing the average point.



Detailed correction · Professor edition

$\bar{x}=3{,}5$ and $\bar{y}=19$,The cloud shows a increasing trend of approximately affine.
Verbate the data that triggers the choice of method.

Trace written

A cloud of dots represents couples (x_i, y_i). The axes, units and scales shall be indicated ; each couple gives only one point.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

$\bar{x} = 3{,}5$ and $\bar{y} = 19$,The cloud shows a increasing trend approximately Affine.

Training · essential

1. For $y = 3x + 13$, estimate y where $x = 8$.
2. Resolve 3 $x + 13 = 50$.
3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Detailed correction · Professor edition

For $x = 8$, $y = 37$. The equation gives $x = \frac{37}{3}$. $x = 8$ gives an interpolation; $x = 30$ one extrapolation.

Reference in the manual

Supplements: Calao, courses p. 110·113; exercises p. 114, n° 1 to 6.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 114, exercise 2

Represent the cloud of points in the next statistical series in a plan coordinate system.

x_i	-3	0	3	5	8
y_i	1	-1	-2	5	7

Detailed correction · Professor edition

Place Points $(-3;1)$, $(0;-1)$, $(3;-2)$, $(5;5)$ and $(8;7)$ in a graduated and leged coordinate system. Check each couple before possibly connecting a trend: a cloud of points is not a Broken line.

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Session 32 — Reading and interpreting the cloud

Two-variate statistics

28/01/2027 • B : 1 h – C : 2 h

Objectives of the session

· Describe a trend.

· Identify dispersal and atypical.

· Avoid abusive causality.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x + 7 - x + 2$. $2x + 9$
2. Calculate the average of 7, 9 and 11. 9
3. Give the slope between $(1, 3)$ and $(4, 9)$. $\frac{9-3}{4-1} = 2$
4. With $y = 2x + 5$, calculate y for $x = 7$. 19
5. Resolve 3 $x + 4 = 25$. $x = 7$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: describe a trend and identify dispersion and Atypical.

Research activity

Compare three clouds: tight increasing trend, dispersed decreasing trend, absence of Write two sentences for each without using the word "cause".

Detailed correction · Professor edition

The description is about direction, form, dispersion and atypical. A cloud does not allow it only to establish a causality. Verbate the data that triggers the choice of method.

Trace written

Describe a cloud consists of commenting on direction, form, dispersion and atypical values, without asserting an unestablished causality.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

The description is about direction, form, dispersion and atypical. A cloud does not allow it only to establish causality.

Training · essential

1. For $y = 4x + 14$, estimate y where $x = 8$.
2. Resolve 4 $x + 14 = 50$.
3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Detailed correction · Professor edition

For $x = 8$, $y = 46$. The equation gives $x = \frac{36}{4}$. $x = 8$ gives an interpolation; $x = 30$ one extrapolation.

Reference in the manual

Supplements: Calao, p. 116, Nos. 11 to 15. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 114, exercise 6

On the chart of the manual is shown the number of cigarettes sold in France according to the number of wolf packs recorded in France between 1995 and 2017.
How can we describe the evolution of cigarette sales from this graph?
So much, can we think that there is a connection between these facts?

Detailed correction · Professor edition

The cloud suggests a decreasing trend: as the number of packs recorded increases, the number of cigarettes sold is decreasing, but this correlation does not allow for a causal link between the two phenomena; a temporal variable or other factors explain their concomitant evolution.

Personal work

Resuming Exercise 3 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation,

To ask the fast students to produce a counter example, justification or a second method.

Meeting 33 — Linear fit

Two-variate statistics

02/02/2027 • B : 2 h – C : 1 h

Objectives of the session

· Get an affine adjustment.

· Interpret a and b.

· Check with the average point.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 20 per cent of 160. 32

2. Give the slope between $(1,3)$ and $(4,9)$. $\frac{9-3}{4-1} = 2$

3. With $y = 2x + 5$, calculate y for $x = 7$. 19

4. Resolve 3 $x + 4 = 25$. $x = 7$

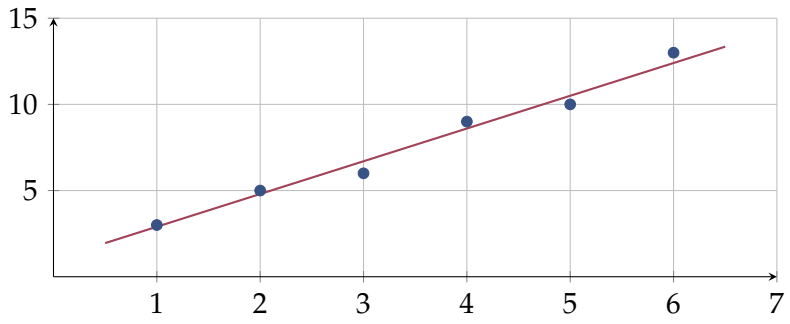
5. Calculate the average of 8, 10 and 12. 10

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: get a fine tuning and interpret a and b.

Research activity

With data from session 31, get regression refines to the calculator. Interpret slope and verify that the average point belongs approximately to the right.



Detailed correction · Professor edition

A regression gives about $y = 3{,}17x + 8{,}57$. An additional EUR 100 000 is associated to about 317 additional sales if y is in hundreds. Make verbalize the data that triggers the choice of method.

Trace written

A Right $y = ax + b$ Adjust a cloud. a represents the estimated average change in y for one additional unit of x.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

A regression gives about $y = 3{,}17x + 8{,}57$. An additional EUR 100 000 is associated to about 317 additional sales if y is in hundreds.

Training · essential

1. For $y = 5x + 15$, estimate y where $x = 8$.
2. Resolve $5 \quad x + 15 = 50$.
3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Detailed correction · Professor edition

For $x = 8$, $y = 55$. The equation gives $x = \frac{35}{5}$. $x = 8$ gives an interpolation; $x = 30$ one extrapolation.

Reference in the manual

Supplements: Calao, courses p. 110·113; exercises p. 117, n° 16 to 18.
session is identified below by its page and number; the reference also allows you to find the figures and digital resources of the manual.

Manuel Calao · p. 115, exercise 9

1. Represent in a plan coordinate system the point cloud associated with the statistical series opposite.

x_i	-2	-1	0	2	3	5
y_i	4	8	12	16	16	20

Draw the right of equation $y = 2x + 10$ in this coordinate system. The adjustment of the cloud by this right Is it satisfactory? If not, propose another one whose equation will be specified reduced.

2. Using this equation, determine:

· the order of the point of the right of adjustment having for abscisse 1 ;

· the order of the point of the right of adjustment having for abscisse 10;

· abscisse from the point of the right of adjustment having as an order 7.

Detailed correction · Professor edition

Right $y = 2x + 10$ correctly follows the general trend of the cloud. $y = 12$ for $x = 1$, $y = 30$ for $x = 10$, and $x = -1{,}5$ if $y = 7$. Value $x = 10$ is extrapolated and should be interpreted with caution.

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 34 — Interpolation

Two-variate statistics

04/02/2027 • B : 1 h – C : 2 h

Objectives of the session

· Interpolate.

· Solve with the model.

· Express an estimate.

Fluency · 6 to 8 minutes, without calculator

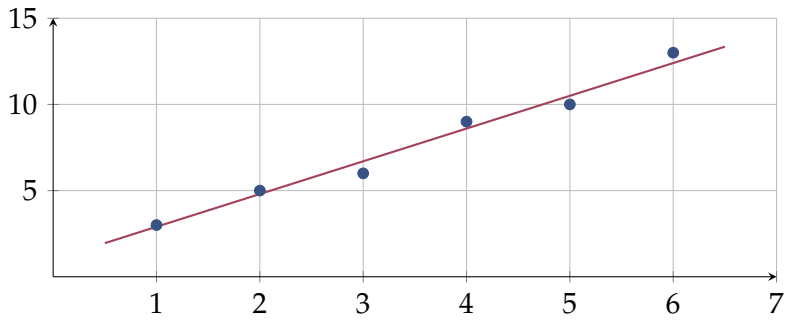
1. Write the multiplier coefficient with an increase of 25 per cent. $1 + \frac{25}{100} = 1,25$
2. With $y = 2x + 5$, calculate y for $x = 7$. 19
3. Resolve 3 $x + 4 = 25$. $x = 7$
4. Calculate the average of 9, 11 and 13. 11
5. Give the slope between $(1, 3)$ and $(4, 9)$. $\frac{9-3}{4-1} = 2$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today.
Before the activity, reformulate the keywords: interpolate and resolve with the model.

Research activity

Over the period observed $x \in [1, 12]$ the model is $y = 2{,}8x + 14$. Estimate y for $x = 7{,}5$, then estimate x where $y = 35$. Qualify each estimate.



Detailed correction · Professor edition

$y = 35$ and $x = 7.5$. The two values are within the observed range: interpolations.
verbalize the data that triggers the choice of method.

Trace written

An interpolation estimates a value within the observed interval.
model-dependent estimate.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

$y = 35$ and $x = 7.5$. Both values are within the observed range: interpolations.

Training · essential

1. For $y = 6x + 16$, estimate y where $x = 8$.
2. Resolve 6 $x + 16 = 50$.
3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Detailed correction · Professor edition

For $x = 8$, $y = 64$. The equation gives $x = \frac{34}{6}$. $x = 8$ gives an interpolation; $x = 30$ one extrapolation.

Reference in the manual

Supplements: Calao, p. 118, n° 19 to 24. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 115, exercise 8

The table below shows the number of new electric vehicles registered in France. 2014 à 2018.

Year		2014	2015	2016	2017	2018
Rank of year	x_i	0	1	2	3	4
Thousands of vehicles	y_i	10,5	17,3	21,7	24,9	31

1. Represent in a coordinate system the point cloud associated with this time series.
2. Propose a fine tuning of this cloud.
3. If it is assumed that this adjustment model remains valid for several years, how many vehicles would have been registered in 2020? From which year would it have exceeded 50,000 New vehicles registered?

Detailed correction · Professor edition

Linear regression gives about $y = 4{,}86x + 11{,}36$. For 2020, $x = 6$, where $y \approx 40{,}5$ or approximately 40,500 vehicles. $y > 50$ corresponds to $x > 7{,}95$: According to this model, it would be crossed in 2022. This is extrapolations.

Personal work

Resuming Exercise 3 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Session 35 — Extrapolation

Two-variate statistics

09/02/2027 • B : 2 h – C : 1 h

Objectives of the session

· Extrapolate carefully.

· Evaluate plausibility.

· Present a model boundary.

Fluency · 6 to 8 minutes, without calculator

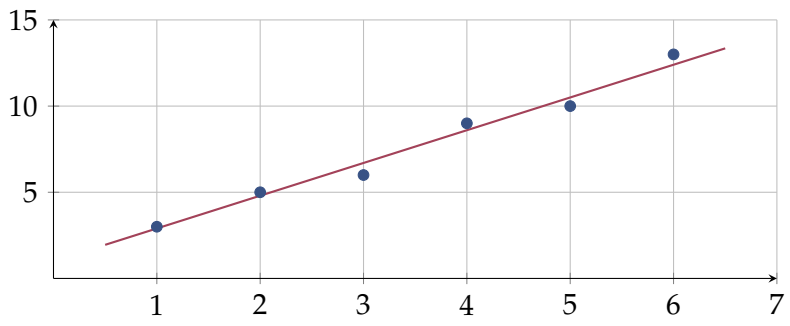
1. Reduce 3 $x + 3 - x + 2$. $2x + 5$
2. Resolve 3 $x + 4 = 25$. $x = 7$
3. Calculate the average of 3, 5 and 7. 5
4. Give the slope between $(1, 3)$ and $(4, 9)$. $\frac{9-3}{4-1} = 2$
5. With $y = 2x + 5$, calculate y for $x = 7$. 19

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: carefully extrapolate and evaluate plausibility.

Research activity

The same model $y = 2{,}8x + 14$, adjusted on $[1, 12]$, is used for $x = 40$. Calculate the estimate, then give three reasons why it can be unreliable.



Detailed correction · Professor edition

The estimate is 126. It is far from the data, the trend can change and constraints the actual variable can be capped. Verbate the data that triggers the choice of method.

Trace written

Extrapolation is estimated to be outside the observed range.
necessary.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

The estimate is 126. It is far from the data, the trend can change and constraints actual can cap the variable.

Training · essential

1. For $y = 2x + 12$, estimate y where $x = 8$.
2. Resolve 2 $x + 12 = 50$.
3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Detailed correction · Professor edition

For $x = 8$, $y = 28$. The equation gives $x = \frac{38}{2}$. $x = 8$ gives an interpolation; $x = 30$ one extrapolation.

Reference in the manual

Supplements: Calao, p. 119, Nos. 25 to 27. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 118, exercise 20

The annual turnover of an enterprise (in EUR '000) was increased during its five years of operation. the first years of existence.

Year		2016	2017	2018	2019	2020
Rank of year	x_i	0	1	2	3	4
Turnover	y_i	315,6	489,9	579,3	763,5	874,5

1. Representing, in an orthogonal coordinate system, the cloud of points $M_i(x_i; y_i)$ of this statistical series.
2. We adjust this cloud of points from the right (M_1M_5). Determine an equation of this line (round the coefficients to the tenth).
3. If the trend continues, what turnover can be expected for 2024? What year will the turnover exceed ·2 million?

Detailed correction · Professor edition

The right through $(0; 315{,}6)$ and $(4; 874{,}5)$ a for slope 139 ,725, i.e. the approximated equation $y = 139{,}7x + 315{,}6$. In 2024, $x = 8$ and $y \approx 1\,433{,}2$, i.e. about 1 ,EUR 433 million. exceeded for first row integer $x = 13$, i.e. in 2029.

Personal work

Resuming Exercise 2 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Meeting 36 — Change of variable

Two-variate statistics

11/02/2027 • B : 1 h – C : 2 h

Objectives of the session

· Make a variable change.

· Adjust new data.

· Return to the initial variable.

Fluency · 6 to 8 minutes, without calculator

1. Calculate 5 % of 80. 4
2. Calculate the average of 4, 6 and 8. 6
3. Give the slope between $(1,3)$ and $(4,9)$. $\dfrac{9-3}{4-1}=2$
4. With $y=2x+5$, calculate y for $x=7$. 19
5. Resolve 3 $x+4=25$. $x=7$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: make a change of variable and adjust the New data.

Research activity

The data seems exponential. $z=\log(y)$ and adjustment gives $z=0{,}08x+1{,}2$. Express y according to the x, then calculate y for $x=5$.

Detailed correction · Professor edition

$y=10^{0{,}08x+1{,}2}$. For $x=5$, $y=10^{1{,}6}\approx 39{,}8$. Have the data that triggers the choice of method.

Trace written

A variable change can align a cloud. After adjustment in the new variable, it must go back to the initial variable.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

$y=10^{0{,}08x+1{,}2}$. For $x=5$, $y=10^{1{,}6}\approx 39{,}8$.

Training · essential

1. For $y = 3x + 13$, estimate y where $x = 8$.
2. Resolve 3 $x + 13 = 50$.
3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Detailed correction · Professor edition

For $x = 8$, $y = 37$. The equation gives $x = \frac{37}{3}$. $x = 8$ gives an interpolation; $x = 30$ one extrapolation.

Reference in the manual

Supplements: Calao, p. 120, n° 28·29. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find the figures and the digital resources of the manual.

Manuel Calao · p. 120, exercise 29

In a concert hall, a reading of sound intensity levels from a power generator
The measurements were recorded in the table below.
below.

Distance x_i (m)	0,5	1	2	3	5	10	11	12	15	20
Sound intensity y_i (dB)	97	85	73	66	59	53	55	52	55	49

An affine adjustment does not seem relevant, a change in variables is envisaged.

1. Complete the table of the manual with $u_i = \log(x_i)$.
2. Represent the dot cloud $(u_i; y_i)$ in a coordinate system, then propose an affine adjustment.
3. Deduce an expression of sound intensity as a function of distance.
4. From the expression found, determine the expected sound intensity at a distance of 8 meters, then 30 meters.

Detailed correction · Professor edition

With the data provided, a regression of y in $u = \log x$ gives approximately $y = -28{,}8u + 83{,}9$. A model is therefore $y(x) = -28{,}8\log(x) + 83{,}9$. It provides approximately 57,9 dB at 8 m and 41.4 dB at 30 m; the second estimate is extrapolation.

Personal work

Resuming Exercise 3 with the following data:
Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Session 37 — Model selection and adjustment

Two-variate statistics

02/03/2027 • B : 2 h – C : 1 h

Objectives of the session

· Compare models.

· Use residues.

· Define a field of use.

Fluency · 6 to 8 minutes, without calculator

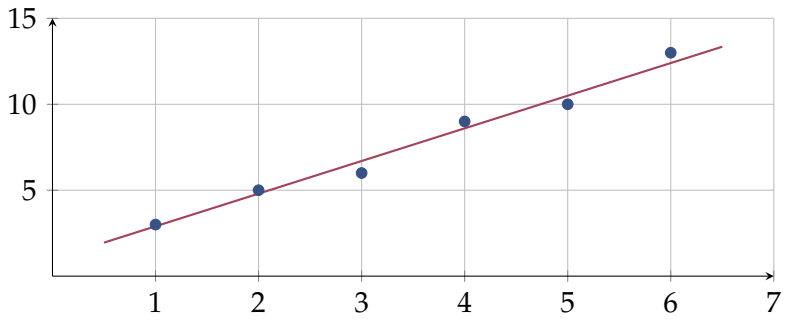
1. Write the multiplier coefficient of a 10 per cent increase. $1 + \frac{10}{100} = 1,10$
2. Give the slope between $(1,3)$ and $(4,9)$. $\frac{9-3}{4-1} = 2$
3. With $y = 2x + 5$, calculate y for $x = 7$. 19
4. Resolve $x + 4 = 25$. $x = 7$
5. Calculate the average of 5, 7 and 9. 7

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: compare models and use residues.

Research activity

Two models give total absolute residues 18 ,4 and 11,7 on the observed area. However, there is an unrealistic growth after $x = 20$. Choosing a model to interpolate $x = 12$, then discuss extrapolation to $x = 50$.



Detailed correction · Professor edition

The second is preferable to interpolate on the observed area if the other criteria are comparable ; no automatic choice is justified for $x = 50$. Making the data that triggers the choice of method.

Trace written

A model is chosen according to the shape of the cloud, the size of the residues and the field of use. best visual fit does not allow unlimited extrapolation.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

The second is preferable to interpolate on the observed area if the other criteria are comparable ; no automatic choice is justified for $x = 50$.

Training · essential

1. For $y = 4x + 14$, estimate y where $x = 8$.
2. Resolve 4 $x + 14 = 50$.
3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Detailed correction · Professor edition

For $x = 8$, $y = 46$. The equation gives $x = \frac{36}{4}$. $x = 8$ gives an interpolation; $x = 30$ one extrapolation.

Reference in the manual

Supplements: Calao, p. 121, No. 30 to 32, and p. 122, No. 33.
is identified below by its page and number; the reference also allows to find the figures and numerical resources of the manual.

Manuel Calao · p. 119, exercise 25

The determination of the caffeine concentration of a drink by measure of absorbance is sought. This, we use a laser beam and we measure the attenuation of the light emitted after crossing the liquid. For the purposes of the experiment, several solutions of which the caffeine content is known are tested and the following set of values is obtained.

Absorbance A	0,155	0,320	0,465	0,630	0,935
Concentration C (mg/L)	5	10	15	20	30

1. Represent this data using a cloud of points in a plan coordinate system.
2. Does an adjustment seem relevant here? If applicable, express the value C of the caffeine concentration as a function of value A Absorbance.
3. For a beverage containing caffeine an absorbance of 0 is measured. ,5. What then is its Caffeine concentration?

Detailed correction · Professor edition

The points are almost aligned. An affine regression gives about $C = 32{,}1A - 0{,}1$. For $A = 0.5$, we get $C \approx 16.0$ mg/L. This estimate is interpolation.

Personal work

Resuming Exercise 2 with the following data:
305 and rate or common difference / ratio by 3. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate confidence level of 1 to 4. Key question(s): 1, 2 and 3.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.

Session 38

Assessment and evaluation

Two-variate statistics

04/03/2027 • B : 1 h – C : 2 h

Objectives of the session

· Choose and operate an adjustment.

· Write a statistical conclusion.

· Analyze an error.

Fluency · 6 to 8 minutes, without calculator

1. Reduce 3 $x + 6 - x + 2$. $2x + 8$
2. With $y = 2x + 5$, calculate y for $x = 7$. 19
3. Resolve 3 $x + 4 = 25$. $x = 7$
4. Calculate the average of 6, 8 and 10. 8
5. Give the slope between $(1, 3)$ and $(4, 9)$. $\frac{9-3}{4-1} = 2$

Reminder and Prerequisites

Read the last written record and know how to perform the elementary calculations mobilized today. Before the activity, reformulate the keywords: choose and exploit an adjustment and write a Statistical conclusion.

Research activity

From a cloud provided by the teacher, produce a complete answer: choice of model, Equation, estimate, type of estimate, unit and reserve. Exchange copies and use grid of six criteria.

Detailed correction · Professor edition

The correction checks the consistency of the model and the presence of the six elements requested, not Only the numerical result. Verbate the data that triggers the choice of method.

Trace written

A complete statistical response contains the model, calculation, rounding, unit, type of estimate and a reservation on validity.

Worked example

Retrieve activity data and write the solution in order: definition of objects, property used, exact calculation or approach, then response phrase. The model response figure in the teaching edition.

Detailed correction · Professor edition

The correction checks the consistency of the model and the presence of the six elements requested, not only the numerical result.

Training · essential

1. For $y = 5x + 15$, estimate y where $x = 8$.
2. Resolve 5 $x + 15 = 50$.
3. The model was drawn up on $[2, 12]$: qualify the estimates for $x = 8$ and $x = 30$.

Detailed correction · Professor edition

For $x = 8$, $y = 55$. The equation gives $x = \frac{35}{5}$. $x = 8$ gives an interpolation; $x = 30$ one extrapolation.

Reference in the manual

Supplements: Calao, p. 122·123. The exercise reproduced in this session is identified below by its page and number; the reference also makes it possible to find figures and resources Numbers of the manual.

Manuel Calao · p. 118·119, exercise 24

Choose the right proposal. Justify the answer. We consider the cloud of points $M_1(1;16)$, $M_2(3;13)$ and $M_3(5;4)$.

1. The average point of this cloud is:
   a. the point of contact $(3;10)$;
   b. the point M_2 ;
   c. the point of contact $(3;11)$.
2. The Right (M_1M_3) a for equation:
   a. $y = 4x + 16$;
   b. $y = -3x + 19$;
   c. $y = -2{,}93x + 17{,}7$.
3. The right of adjustment obtained by the calculator:
   a. go through one of the three points;
   b. is parallel to the right (M_1M_3) ;
   c. goes through the origin of the coordinate system.

Detailed correction · Professor edition

1. Response c: $= 3$ and $= 11$.
2. Answer b: the slope is $(4 - 16)/(5 - 1) = -3$ and the original order is 19.
3. Answer b: regression is $y = -3x + 20$, it passes through the middle point and is parallel to (M_1M_3).

Personal work

Resuming Exercise 3 with the following data:
310 and rate or common difference / ratio by 4. Write in four lines: data, formula, calculation, conclusion.

Detailed correction · Professor edition

The correction depends on the exercise taken over. Check above all the consistency of the model, the number Iterations, rounding and the conclusion sentence.

Exit ticket · 3 minutes

On half a sheet: write the main rule of the session, deal with a very short example and indicate the degree of confidence from 1 to 4. Key question(s): 1 and 2.

Method

Drive in 1 hour: 7 min of automatics; 10 min of pooling and trace; 28 min of exercises essentials ; 5 min correction ; 3 min exit ticket.
Drive in 2 hours: add 20 min of activity research, 20 min of institutionalization, one second autonomous exercise and 10 minutes of comparative correction.
Differentiation: provide fragile students with a four-box card (data, model, calculation, To ask the fast students to produce a counter example, justification or a second method.